

AMHERST, NS, Canada

WMO: 743905

Lat: 45.8509N

Lon: 64.2605W

Elev: 75

StdP: 14.66

Time Zone: -4.00 (NAA)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-1.6	-23.5	1.5	-5.4	-17.1	2.2	-0.5	36.4	38.0	32.1	28.4	12.9	290	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	77.9	67.1	75.3	66.2	73.0	65.0	69.8	75.0	68.0	72.5	66.5	70.8	10.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	102.5	72.7	66.2	96.9	70.8	64.9	92.3	69.3	33.8	75.3	32.4	72.6	31.2	71.0	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.4	25.7	23.0	-12.5	83.9	5.8	2.3	-16.7	85.6	-20.1	86.9	-23.4	88.2	-27.6	89.9		
(5)			-13.5	73.4	5.6	1.6	-17.5	74.5	-20.7	75.5	-23.9	76.4	-27.9	77.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	18.2	20.6	28.7	38.8	49.3	58.2	64.7	64.4	58.0	47.2	37.6	26.9		
	DBStd	17.58	11.13	10.26	8.00	6.18	5.69	5.44	3.15	4.06	6.01	6.58	8.26	9.07		
	HDD50	4133	985	822	659	341	82	4	0	0	8	133	383	716		
	HDD65	8181	1450	1242	1124	787	487	210	43	61	222	551	823	1181		
	CDD50	1524	0	0	0	4	61	252	457	446	246	48	10	0		
	CDD65	95	0	0	0	0	0	7	35	42	11	0	0	0		
	CDH74	356	0	0	0	0	9	42	133	135	37	0	0	0		
	CDH80	23	0	0	0	0	0	3	9	7	4	0	0	0		
	WSAvg	10.5	11.1	11.1	11.2	10.9	10.3	9.8	9.8	9.2	9.4	10.5	11.0	11.3		
	PrecAvg	47.7	4.4	3.7	4.0	3.7	3.6	3.5	3.9	3.5	3.4	4.3	4.3	5.1		
	PrecMax	68.7	11.6	5.8	6.5	9.2	9.6	7.3	7.0	7.4	6.7	7.8	8.4	13.0		
	PrecMin	31.4	0.9	1.3	1.1	1.0	0.7	1.0	0.7	0.2	0.4	1.9	1.5	2.0		
	PrecStd	8.3	2.2	1.1	1.3	1.8	1.9	1.5	1.9	1.9	1.5	1.8	1.6	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	48.1	52.8	62.2	73.6	79.4	81.2	80.8	78.3	68.5	64.2	53.4		
		MCWB	50.2	46.9	49.9	53.8	61.8	67.7	68.7	67.1	67.5	63.4	61.1	51.8		
	2%	DB	44.1	41.5	47.5	56.1	67.4	74.5	77.6	77.5	73.6	64.4	59.9	48.4		
		MCWB	42.9	40.2	44.7	50.3	58.3	64.4	67.2	67.4	65.8	59.7	57.5	47.1		
	5%	DB	38.1	37.9	43.3	52.2	63.9	71.1	74.9	75.0	70.6	60.9	54.1	43.3		
		MCWB	36.3	36.1	40.5	45.8	55.5	62.1	66.0	66.6	64.3	56.4	51.7	41.1		
	10%	DB	33.6	35.2	39.8	48.9	60.1	68.3	72.6	72.5	67.8	58.0	49.2	39.0		
		MCWB	31.9	33.5	37.3	43.6	52.9	61.0	65.1	65.1	62.3	54.4	46.7	36.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	47.0	51.1	57.1	63.5	70.5	71.8	72.4	70.4	64.6	61.8	52.1		
		MCDB	51.4	47.8	52.1	61.6	70.4	76.5	77.7	76.9	76.2	68.0	63.9	52.7		
	2%	WB	43.2	40.6	45.5	52.2	60.3	66.6	69.5	69.7	67.5	61.1	57.7	47.2		
		MCDB	44.4	41.3	47.1	55.6	65.8	71.5	74.9	74.4	71.3	64.0	59.9	48.4		
	5%	WB	36.7	36.6	41.1	47.4	57.3	64.5	67.7	67.9	65.7	57.5	52.4	41.6		
		MCDB	37.9	37.7	42.9	50.9	62.0	69.4	72.3	72.4	69.1	59.7	54.2	42.7		
	10%	WB	32.2	33.8	37.6	44.5	54.2	62.4	66.2	66.4	63.3	54.7	47.5	37.0		
		MCDB	33.4	35.2	39.7	48.2	58.8	66.5	70.8	70.6	66.8	57.5	49.3	38.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	15.7	13.3	14.1	17.1	17.4	15.9	15.9	16.3	14.8	12.3	13.4		
		MCDBR	23.5	16.1	18.0	20.5	23.4	21.2	20.0	19.0	18.7	18.3	17.2	17.6		
		MCWBR	24.3	15.9	16.2	15.1	15.2	13.1	11.9	10.7	11.8	14.1	17.0	17.7		
	5% WB	MCDBR	24.7	15.4	18.2	18.4	20.5	17.4	16.5	15.0	15.5	16.8	16.9	17.7		
		MCWBR	25.8	15.5	17.9	15.5	14.6	12.6	11.0	9.8	11.1	15.0	17.0	18.4		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.288	0.323	0.349	0.372	0.388	0.416	0.395	0.357	0.327	0.306	0.289		
	taud		2.408	2.419	2.405	2.407	2.389	2.360	2.317	2.367	2.464	2.541	2.536	2.493		
	Ebn at Noon		257	285	288	289	284	279	269	271	273	267	254	249		
	Edh at Noon		23	27	32	35	37	38	39	36	30	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		442	692	1071	1345	1577	1708	1751	1559	1234	782	433	329		
	RadStd		38	60	72	127	164	140	118	107	99	81	49	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.70	N/A	N/A	N/A	N/A

Nomenclature: See separate page